

SENDING DATA TO THINGSPEAK: Thingspeak Channel Setup Recap

Ts. Dr. Razif bin Razali

CREATE NEW CHANNEL

ThingSpeak™

My Channels

New Channel

Search by tag



Help

Collect data in a ThingSpeak channel from a device, from another channel, or from the web.

Click **New Channel** to create a new ThingSpeak channel.

Click on the column headers of the table to sort by the entries in that column or click on a tag to show channels with that tag.

Learn to [create channels](#), explore and transform data.

Learn more about [ThingSpeak Channels](#).

Examples

- [Arduino](#)
- [Arduino MKR1000](#)
- [ESP8266](#)
- [Raspberry Pi](#)
- [Netduino Plus](#)

Upgrade

Need to send more data faster?

Need to use ThingSpeak for a commercial project?

Upgrade

ThingSpeak™

New Channel

Name

TEMP & HUMIDITY

Description

RECORD TEMP AND HUMIDITY

Field 1

TEMP



Field 2

HUMIDITY



Field 3



Field 4



Video URL

http://

Show Status



Save Channel

Help

Channels store all the data that a ThingSpeak application collects. Each channel includes eight fields that can hold any type of data, plus three fields for location data and one for status data. Once you collect data in a channel, you can use ThingSpeak apps to analyze and visualize it.

Channel Settings

- **Percentage complete:** Calculated based on data entered into the various fields of a channel. Enter the name, description, location, URL, video, and tags to complete your channel.
- **Channel Name:** Enter a unique name for the ThingSpeak channel.
- **Description:** Enter a description of the ThingSpeak channel.
- **Field#:** Check the box to enable the field, and enter a field name. Each ThingSpeak channel has eight fields.

ThingSpeak™

TEMP & HUMIDITY

Channel ID: **2978383**

Author:

[mwa000037889969](#)

Access: Private

RECORD TEMP AND HUMIDITY

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

+ Add Visualizations

+ Add Widgets

Export recent data

MATLAB Analysis

MATLAB Visualization

Channel Stats

Created: [less than a minute ago](#)

Entries: 0

Field 1 Chart



TEMP & HUMIDITY

PRIVATE & PUBLIC VIEW OF THINGSPEAK



Channels ▾ Apps ▾ Devices ▾ Support ▾

Commercial Use How to Buy 

TEMP & HUMIDITY

Channel ID: 2978383

Author: mwa0000037889969

Access: Public

RECORD TEMP AND HUMIDITY

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

Channel Sharing Settings

- ☐ Keep channel view private
- ☒ Share channel view with everyone
- ☐ Share channel view only with the following users:

Email Address

Enter email here

Add User

Help

ThingSpeak allows you to control who can view the data in your channel. Irrespective of the settings on this tab, reading data from or writing data to the fields of a channel requires the appropriate API key for the channel.

Channel Sharing Settings

- **Keep channel view private:** Selecting this option keeps your channel private. Only you will be able to see the channel view.
- **Share channel view with everyone:** Selecting this option makes the public view of your channel viewable by anyone browsing the ThingSpeak website.
- **Share channel view only with the following users:** Selecting this option shares the private view of your channel only with specific ThingSpeak users.

UNDERSTANDING CHANNEL ID & API KEYS

Channel Information

- **TEMP & HUMIDITY:** This is the **Name** of the current ThingSpeak Channel. It's user-defined and describes the type of data being collected.
- **Channel ID: 2978383:** This is a unique numerical identifier for this specific channel. You'll use this ID in your device code or API requests to specify which channel to interact with.
- **Author: mwA0000037889969:** The username or identifier of the ThingSpeak user who created or owns this channel.
- **Access: Public:** This indicates that the data in this channel (specifically the "Public View") can be viewed by anyone without needing an API key. If it were "Private," only the owner or those with a Read API Key could see the data.
- **RECORD TEMP AND HUMIDITY:** This appears to be a **Description** for the channel, providing more context about its purpose.

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Commercial Use How to Buy WA

TEMP & HUMIDITY

Channel ID: 2978383
Author: mwA0000037889969
Access: Public

RECORD TEMP AND HUMIDITY

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

Write API Key

KeyCUPRHU8PH0VRD6SM

Generate New Write API Key

Read API Keys

KeyYB1ECI9QJBJOAYH

Note

Save NoteDelete API Key

Add New Read API Key

Help

API keys enable you to write data to a channel or read data from a private channel. API keys are auto-generated when you create a new channel.

API Keys Settings

- **Write API Key:** Use this key to write data to a channel. If you feel your key has been compromised, click **Generate New Write API Key**.
- **Read API Keys:** Use this key to allow other people to view your private channel feeds and charts. Click **Generate New Read API Key** to generate an additional read key for the channel.
- **Note:** Use this field to enter information about channel read keys. For example, add notes to keep track of users with access to your channel.

API Requests

Write a Channel Feed

GET https://api.thingspeak.com/update?api_key=CUPRHU8PH0VRD6SM&field1=0

Read a Channel Feed

GET https://api.thingspeak.com/channels/2978383/feeds.json?results=2

Read a Channel Field

GET https://api.thingspeak.com/channels/2978383/fields/1.json?results=2

Read Channel Status Updates

GET https://api.thingspeak.com/channels/2978383/status.json

Learn More

UNDERSTANDING CHANNEL ID & API KEYS

Write API Key:

- **Key: CUPRHU8PH0VRD6SM:** This is a secret alphanumeric string. **This key allows you or your IoT device to send data to this ThingSpeak channel.** It's like a password for writing data. You embed this key in the code of your sensor device (e.g., an ESP32, Arduino) so it can update the channel fields.

Read API Keys:

- **Key: YB1ECI9QJBJOAOYH:** This is also a secret alphanumeric string. **This key allows you or applications to read data from this ThingSpeak channel, especially if the channel is private.** If the channel is public, you might not always need a Read API key to fetch data,

The screenshot displays the ThingSpeak interface for channel 2978383, titled "TEMP & HUMIDITY". The channel is public, with author "mwa000037889969". The "API Keys" tab is selected, showing a "Write API Key" section with the key "CUPRHU8PH0VRD6SM" and a "Generate New Write API Key" button. Below this is a "Read API Keys" section with the key "YB1ECI9QJBJOAOYH", a "Note" field, and "Save Note" and "Delete API Key" buttons. An "Add New Read API Key" button is at the bottom. On the right, a "Help" section explains API keys and lists settings. Below that, "API Requests" are shown with example GET URLs for updating, reading feeds, reading fields, and checking status.

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TEMP & HUMIDITY

Channel ID: 2978383
Author: mwa000037889969
Access: Public

RECORD TEMP AND HUMIDITY

Private View Public View Channel Settings Sharing API Keys Data Import / Export

Write API Key

Key CUPRHU8PH0VRD6SM

Generate New Write API Key

Read API Keys

Key YB1ECI9QJBJOAOYH

Note

Save Note Delete API Key

Add New Read API Key

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API Requests

Write a Channel Feed

GET https://api.thingspeak.com/update?api_key=CUPRHU8PH0VRD6SM&field1=0

Read a Channel Feed

GET <https://api.thingspeak.com/channels/2978383/feeds.json?results=2>

Read a Channel Field

GET <https://api.thingspeak.com/channels/2978383/fields/1.json?results=2>

Read Channel Status Updates

GET <https://api.thingspeak.com/channels/2978383/status.json>

[Learn More](#)

SENDING DATA TO THINGSPEAK: HTTP Basics for IoT

Ts. Dr. Razif bin Razali

SENDING SENSOR DATA TO THINGSPEAK

Install Necessary Libraries:

- Open Arduino IDE. Go to "Sketch" -> "Include Library" -> "Manage Libraries...".
- **DHT sensor library:** Search for "DHT sensor library" by Adafruit. Install it.
- **Adafruit Unified Sensor library:** The DHT library often depends on this. Search for "Adafruit Unified Sensor" by Adafruit. Install it if it's not already installed as a dependency.
- **(For ESP8266) ThingSpeak library:** Search for "ThingSpeak" by MathWorks. Install it.

Select Your Board and Port:

- In Arduino IDE, go to "Tools" -> "Board" and select your specific ESP8266 board (e.g., "NodeMCU 1.0 (ESP-12E Module)") or ESP32 board (e.g., "ESP32 Dev Module").
- Connect your ESP board to your computer via USB.
- Go to "Tools" -> "Port" and select the correct COM port.

Code to test - put this code into github

Refer to github

https://github.com/razifrazali66/IoT/blob/main/04_data2thingSpeak/01_toThingSpeak.txt

HOW IOT DEVICES USE HTTP TO COMMUNICATE

IoT devices act as clients that use HTTP(S) to:

- **POST** sensor readings, events, and status updates (formatted as JSON, XML, etc.) to specific API endpoints on a cloud platform. This data is then stored, processed, or visualized.
- **GET** configuration settings, firmware updates, or commands from the cloud by requesting specific API endpoints.
- **Authenticate** themselves using API keys, tokens, or certificates embedded in the HTTP headers.
- Rely on **HTTPS** to ensure the confidentiality and integrity of the data exchanged.

BRIEF OVERVIEW OF HTTP GET & POST REQUESTS

Feature	HTTP GET	HTTP POST
Purpose	To retrieve or request data from a specified resource.	To send data to a server to create or update a resource.
Data Sending	Data is appended to the URL in a query string (e.g., <code>www.example.com/api/data?id=123&value=42</code>).	Data is sent in the body of the HTTP request, not in the URL.
Data Size	Limited by URL length restrictions (varies by browser/server, typically around 2048 characters).	Can send much larger amounts of data as it's in the request body.
Visibility	Data is visible in the URL, browser history, and server logs. Not suitable for sensitive data.	Data is not visible in the URL. More suitable for sensitive data (though HTTPS is crucial for actual security).
Idempotency	Generally idempotent (making the same request multiple times produces the same result).	Not necessarily idempotent (making the same request multiple times may create multiple resources or have different effects).
Caching	Responses can be cached by browsers and proxies.	Responses are not typically cached unless explicitly configured.
Bookmarking	Can be bookmarked as the full request is in the URL.	Cannot be easily bookmarked with the data payload.
Use Cases	Fetching web pages, images, data; querying information.	Submitting forms, uploading files, creating new records, updating existing data.